ACRES — Automated & Centralized Rental Estate

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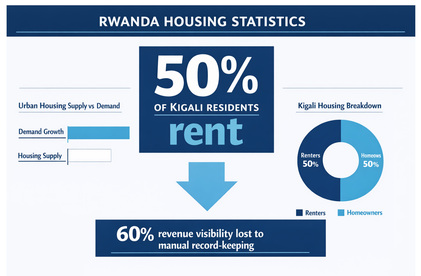
1. Problem, Context, and Users

Modernising rental management to secure housing and financial transparency



Over **50%** of Kigali residents rent, yet landlord–tenant relationships run on paper leases and handwritten receipts. Rwanda's rapid urbanisation is outpacing these systems, exposing landlords to revenue leakage and leaving tenants without proof of payment.

2. Objectives and Scope

- **Aim:** Centralised web platform to digitise property management and lease compliance.
- **Obj. 1:** Multi-unit dashboard for financial and occupancy oversight.
- **Obj. 2:** Secure digital vault for leases with automated invoicing.
- **Obj. 3:** QR-code maintenance reporting requiring zero app installation.
- **In scope:** Lease tracking, payment logging, maintenance ticketing, role-based access.
- **Out of scope:** Bank API integration, IoT smart locks, multi-currency support.

3. Project Metadata

Team	Eelaf Adam, Becky Alek, Prince Niyomugabo, Tumba II Kongolo, Peniel Obeng, Benjamin Kettey-Tagoe
Programme	BSE — Foundations Project / C2
Supervisor	David Neza Tuyishimire
Demo / QR	`Scan QR for live demo & repo`

4. Constraints and Design Decisions

- **Connectivity:** SSR pages optimised for low-bandwidth mobile networks.
- **Legal:** Complies with Rwanda's 2021 Data Protection Act; PII encrypted via bcrypt & AES-256.
- **Literacy:** One-tap UI workflows designed for varied digital literacy.
- **Stack:** Next.js + Node.js + MySQL for SSR performance and data integrity.
- **Auth:** Unit-specific QR codes — no tenant password required.
- **Trade-off:** Zero-install simplicity chosen over a richer native mobile app.

5. Proposed Solution and System Overview

ACRES — an Administrative Engine turning notebooks into a digital dashboard

Solution summary

ACRES digitalises every landlord–tenant touchpoint. Landlords upload leases, log payments, and monitor occupancy from one dashboard. Tenants scan a door-mounted QR code to submit maintenance requests with no login or app needed. Alerts fire automatically for due rent, expiring leases, and unresolved tickets.

Architecture overview

- **Frontend:** Next.js + Tailwind CSS (SSR)
- **Backend:** Node.js REST API, JWT auth
- **Database:** MySQL — units, leases, payments
- **Auth:** bcrypt hashing; QR-token tenant access
- **Hosting:** Vercel (FE) + Railway (API + DB)

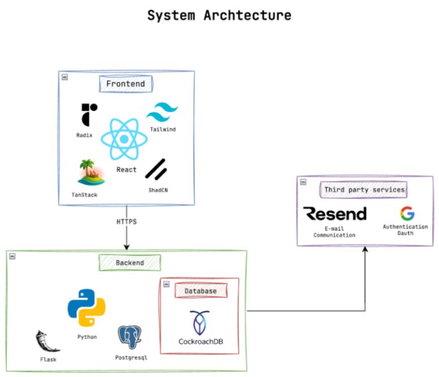


Figure 1: ACRES system architecture and data-flow diagram

6. Methodology and Build Process

Research & Discovery

- **Interviews:** 12 landlords across Bumbogo & Kimironko.
- **Requirements:** Lease storage, payment logging, maintenance ticketing.
- **Compliance:** Data flows mapped against Rwanda's Data Protection Act.

Engineering & Implementation

- **Architecture:** 3-tier Next.js / Node.js / MySQL stack.
- **Prototyping:** QR scanning tested on mid-range Android devices.
- **UAT:** Task scenarios with 12 landlords; two UI iterations.

Stage	Key output
Discover	60% revenue leakage identified in manual tracking
Design	Admin Engine ERD; QR-code maintenance ticketing flow
Build	Next.js SSR, Node.js API, MySQL schema, bcrypt layer
Test	95% task-success rate; avg. response time 1.8 s
Iterate	Simplified dashboard layout post-UAT feedback

7. Results, Evaluation, and Demo Evidence

95%	1.8s	12
Task success	Avg. response	Pilot landlords

Key results

- Payment log time cut to **1.8 s** vs. minutes on paper.
- **100 leases** successfully digitised in pilot.
- **11/12 landlords** rated dashboard "highly intuitive."
- Zero data-loss across all UAT sessions.

Demo capabilities

- End-to-end maintenance cycle (QR scan to resolve).
- Lease Vault live with automated expiry alerts.
- Unit-level financial dashboard operational.
- Payments currently manual input (MoMo pending).

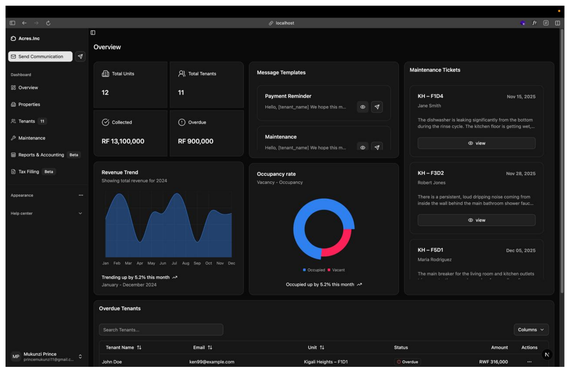


Figure 2: ACRES landlord dashboard — occupancy, payments, and maintenance tickets

8. Impact, Contribution, and Learning

- **Technical:** QR-to-Web workflow enables zero-friction tenant access with full audit trails.
- **Users:** Landlords gain financial visibility; tenants get timestamped receipts and a maintenance pipeline.
- **Research:** UAT confirmed landlords prioritise *legal security* over complex accounting tools.
- **Mission:** Supports Rwanda's National Housing Policy by professionalising informal rentals.
- **Learning:** Deepened skills in SSR optimisation, data-protection compliance, and UCD.

It doesn't just work — it protects tenants and empowers landlords.

9. Limitations and Next Steps

Limitations

- No MTN MoMo / Airtel API — payments verified manually.
- Assumes basic smartphone access and QR-scanning literacy.
- Small pilot (12 landlords); broader validation needed.

Next steps

- Integrate MTN MoMo & Airtel Money for auto-reconciliation.
- Add SMS/USSD fallback for non-smartphone tenants.
- Expand to 50+ landlords across 5 Kigali districts.
- Explore RDB registration for formal property managers.

10. References and Acknowledgements

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Acknowledgements

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Figure 3: ACRES PMS dashboard

LaTeX TikZposter